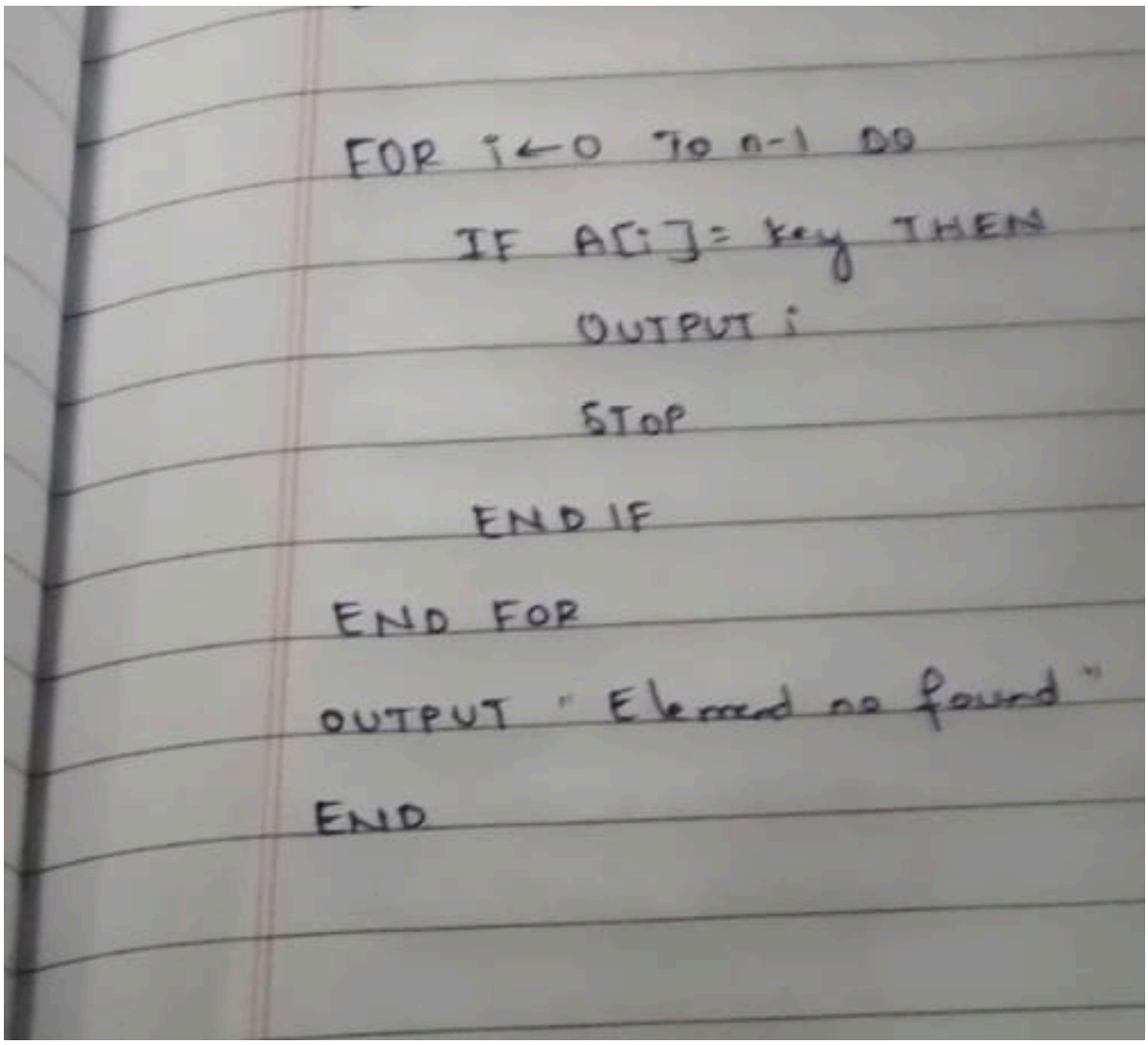
Practical-2	***Implementation and Time analysis of linear and binary search algorithm.***
Date :	

1)Linear search



code:

```
#include <iostream>
using namespace std;

int linearSearch(int arr[], int n, int key) {
    for (int i = 0; i < n; i++) {
        if (arr[i] == key) {
            return i;
        }
    }

    return -1;
}

int main() {
    int arr[] = {10, 20, 30, 40, 50};
    int n = sizeof(arr) / sizeof(arr[0]);

    int key = 30;

    int result = linearSearch(arr, n, key);

    if (result != -1)
        cout << "Element found at index " << result;
    else
        cout << "Element not found";

    return 0;
}
```

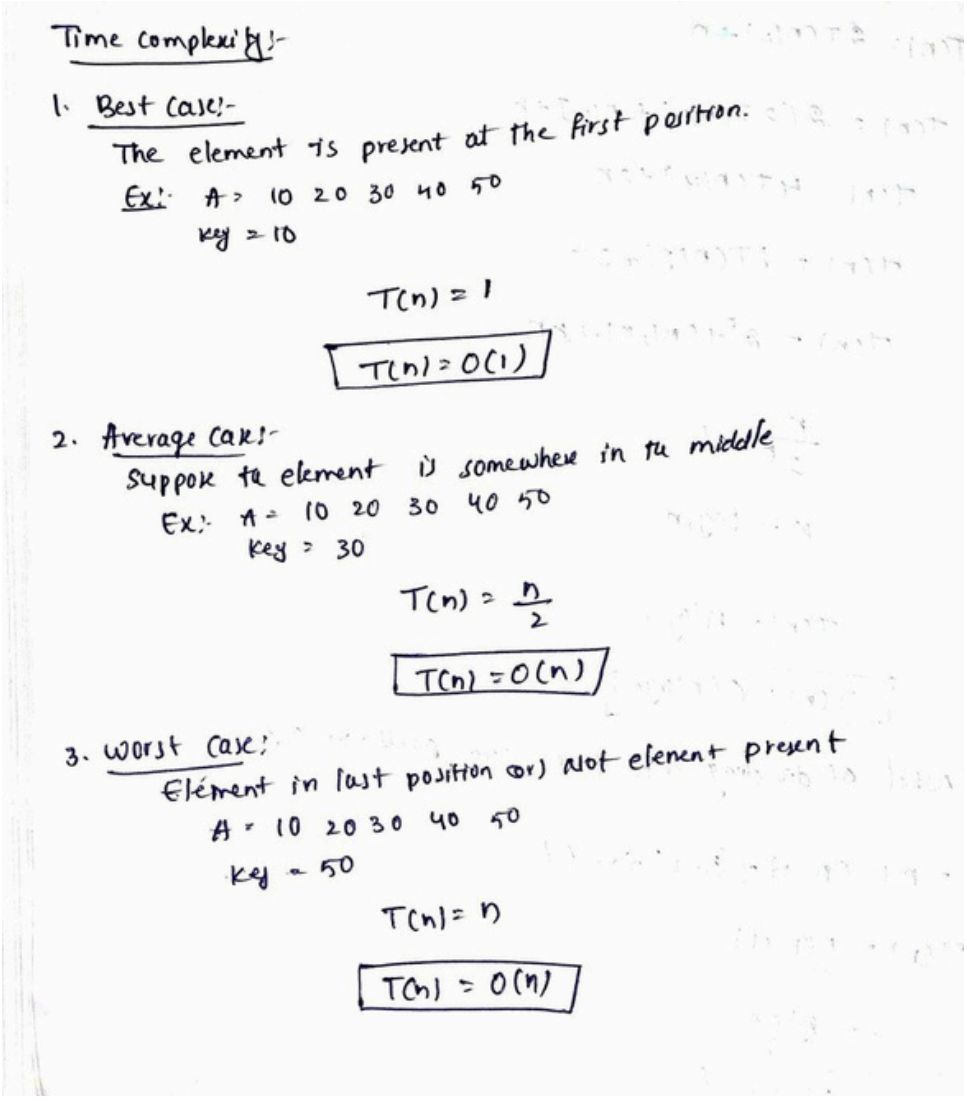
Output:

```
Output

Element found at index 2

=== Code Execution Successful ===
```

Time complexity:



1)Binary search

```
WHILE low ≤ high DO
    mid ← (low + high) / 2
    IF A[mid] = key THEN
      OUTPUT mid
      STOP
    ELSE IF A[mid] < key THEN
      low ← mid + 1
    ELSE
      high ← mid - 1
    ENDIF
END WHILE
OUTPUT " Element not found"
END
```

code:

```
#include <iostream>
using namespace std;

int binarySearch(int arr[], int n, int key) {
    int low = 0;
    int high = n - 1;

    while (low <= high) {
        int mid = (low + high) / 2;

        if (arr[mid] == key) {
            return mid;
        }
        else if (arr[mid] < key) {
            low = mid + 1;
        }
        else {
            high = mid - 1;
        }
    }

    return -1;
}

int main() {
    int arr[] = {10, 20, 30, 40, 50};
    int n = sizeof(arr) / sizeof(arr[0]);

    int key = 40;

    int result = binarySearch(arr, n, key);

    if (result != -1)
        cout << "Element found at index " << result;
    else
        cout << "Element not found";

    return 0;
}
```

Output:

```
Output

Element found at index 3

=== Code Execution Successful ===
```

Time complexity:

1) Best case:-
The element is found at the middle position in the first comparison.
Ex:- A = 10 20 30 40 50
key = 30
A[mid] = key
$T(n) = 1$
$T(n) = O(1)$

2) Average case:-
On Average, the search also eliminates approximately half of the remaining elements at each step.
$T(n) \approx \log_2 n$
$T(n) = O(\log n)$

3) Worst case
This is where Binary search becomes interesting
$n \rightarrow n/2 \rightarrow n/4 \rightarrow \cdots 1 \rightarrow$
$\frac{n}{2^k} = 1$
$n = 2^k$
$k = \log_2 n$
$T(n) = \log n$
$T(n) = O(\log n)$

Conclusion:

In this practical, we implemented Linear Search and Binary Search and understood their working and time complexities. Linear Search checks elements one by one, while Binary Search repeatedly divides a sorted array into halves. Thus, Binary Search is more efficient for large sorted datasets.